

ALFIDO TECH MACHINE LEARNING INTERNSHIP



PROJECT REPORT

Task Name: Iris Classification

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1.INTRODUCTION:

Machine learning is a branch of Artificial Intelligence that enables systems to learn from data and make predictions.

2.OBJECTIVE:

The objective of this project is to classify iris flowers using machine learning algorithms.

3.Tools & Technologies Method:

Python

JupyterNotebook

Pandas

Numpy

Matplotlib

Seaborn

Scikit-learn

4.Dataset Information:

The Iris dataset contains flower measurements such as sepal-length, sepal width, petal length, petal width.

This dataset is the famous Iris Flower Dataset used in Machine Learning and Data Science for classification problems.

Dataset Overview

Total Rows (Records): 150

Total Columns: 5

Type of Dataset: Classification Dataset

Target Column: species

It contains measurements of iris flowers from 3 different species.

Flower Species in Dataset

The dataset contains 3 flower categories:

Iris-setosa

Iris-versicolor

Iris-virginica

Each species has:

50 records

Different flower measurements

Example Record Explanation

Example row from your dataset:

sepal_length

sepal_width

petal_length

petal_width

species

5.1

3.5

1.4

0.2

Iris-setosa

Meaning:

Sepal length = 5.1 cm

Sepal width = 3.5 cm

Petal length = 1.4 cm

Petal width = 0.2 cm

Flower belongs to Iris-setosa

Input and Output

Input Features (X)

These columns are used as inputs:

sepal_length

sepal_width

petal_length

petal_width

Output/Target (Y)

species

5.METHODOLOGY:

1.Imported required libraries

2.Loaded Dataset

3.Performed Data analysis

4.Created graphs

5.Split Dataset

6.Trained model

7.Predicted output

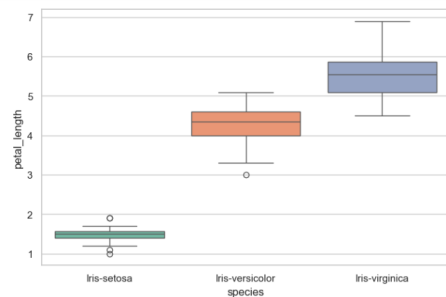
8.Evaluated accuracy

6.DATA VISUALIZATION:

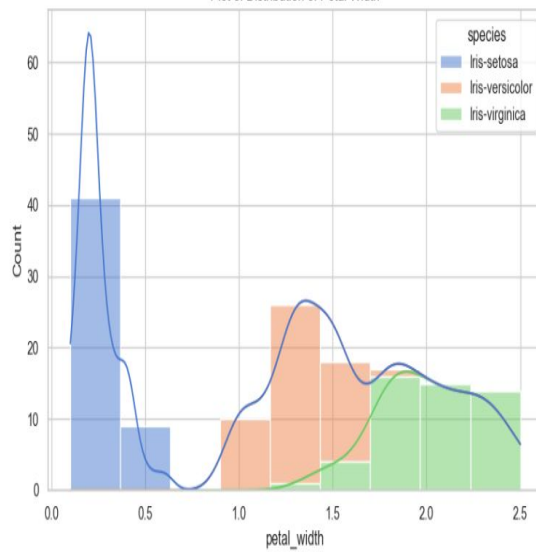
```

best_model_instance = model
print("\n\n>> Selected Best Model: (best_model_name) ((best_accuracy*100:.2f)% Accuracy)")
X_te_final = X_test_scaled if best_model_name != "Decision Tree" else X_test
final_preds = best_model_instance.predict(X_te_final)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, final_preds))
print("\nClassification Report:")
print(classification_report(y_test, final_preds))
joblib.dump(best_model_instance, 'best_iris_model.pkl')
print("Best model saved successfully!")

```



Plot 3: Distribution of Petal Width



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7.MODEL TRAINING:

Logistic Regression algorithm, K-NearestNeighbor, Decision Tree was used to train the model.

8.PREDICTION RESULTS:

The model successfully predicted flower categories.

9.ACCURACY:

The model achieved 93.33% accuracy



10.CONCLUSION:

Successfully completed iris classification using machine learning techniques.